Notes on Data Encryption

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Document is WIP, typos could be found

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- 1 - Introduction.

Data encryption dials with how informations, see [4] for the definition, can be transmited securely through a channel.

More in details, if m is a message we want to send through a channel, first we have to encode it in a form that allows it to be transmited; then we want that such message is meaningless to those who have no right to read it. On the other hand, we want to be able to retrieve the message once this is recived.

To achive the above two disciplines were found: Cryptography and Code theory; the former focuses on how to make data secure, the latter provides methods to transmit data through a channel.

- 1.1 - About these notes.

These notes will focus on both Cryptography and Code theory, or at least, some parts of them.

More precisely, we begin by introducing some fundamental concepts of abstract algebra and projective geometry, we then discuss what public key encryption is focusing on the RSA and the ElGamal algorithm; we continue with describing some algorithms to factor integers, and conclude with several types of codes.

The following notation will be used throughout the document.

- $\mathcal{A}, \mathcal{B}$ and $\mathcal{E}$ will indicate, respectively, the sender, the reciever and an unauthorized individual attempting to capt our message.
- $\mathcal{C}$ will denote codes.
- Words in a code $\mathcal{C}$ will be represented by $\mathbf{w}$, with the occasional use of subscripts.

Any other symbol will be explained beforehand.

- 2 - Mathematical background.

In this section we provide the reader with some notions of abstract algebra; we recall what Greatest Common Divisor (GCD) and Least Common Multiple (LCM) are, we discuss congruence modulo n, and present groups and rings, and discuss some well know results on them.

Also, some broad concepts of projective geometry will be given; we focus on what projective space is, what the sum of projective points looks like, and to conclude, we give consider elliptic curves.

- 2.1 - Some abstract algebra.

In what follows, we assume the reader to have a basic understanding of number sets and equivalence relations.

Definition (Divisibility) Let $a, b \in \mathbb{Z}$. We say that a divides b, and write $a|b$, if

$$\exists q \in \mathbb{Z} \text{ s.t. } b = qa.$$

Otherwise we say that a does not divide b, and write $a \nmid b$.

Divisibility allows us to define two important concepts: LCM and GCD.

Definition (LCM) Let $a, b \in \mathbb{Z}$. Let $m \in \mathbb{Z}$. We say that m is the LCM for a and b if, the following holds:

1. $a|m$ and $b|m$, and
2. If $\exists m' \in \mathbb{Z}$ such that $a|m'$ and $b|m'$, then $m|m'$.

Definition (GCD) Let $a, b \in \mathbb{Z}$. Let $d \in \mathbb{Z}$. We say that d is the GCD for a and b if, the following holds:

1. $d|a$ and $d|b$, and
2. If $\exists d' \in \mathbb{Z}$ such that $d'|a$ and $d'|b$, then $d'|d$.

The computation of the GCD is done by the euclidean algorithm; we present an efficient inpementation of it in *Section* **??**.

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Definition (Congruence modulo n) Let $a, b \in \mathbb{Z}$, and let $n \in \mathbb{Z}$ be a fixed integer. We say that a is congruent b modulo n, and write $a \equiv b \pmod{n}$ if and only if

$$\exists k \in \mathbb{Z} \text{ s.t. } a = kn + b.$$

- 2.1.1 - A brief introduction to groups.

Definition (Group) Let G be a set. Assume $\cdot : G \to G$ to be some function. We say that the pair $(G, \cdot)$, often simply as G, is a group if the following properties hold:

- $\forall a, b, c \in G, a \cdot (b \cdot c) = (a \cdot b) \cdot c$;
- $\exists e \in G \ \forall g \in G$ s.t. $g \cdot e = e \cdot g = g$;
- $\forall g \in G \ \exists g^{-1} \in G$ s.t. $g \cdot g^{-1} = g^{-1} \cdot g = e$.

If it also holds that $\forall a, b \in G, a \cdot b = b \cdot a$, we say that G is an abelian (or commutative) group.

We consider finite groups: that is, group with finitely many elements. Precisely, we are interested on the following class of groups.

Definition (Cyclic group) Let G be a group, and let $g \in G$. Denote by g^k the application of $\cdot$ on g, k times. Denote by $\langle g \rangle$ the set of all powers of g. We say that G is cyclic if $\langle g \rangle = G$.

- 2.1.2 - A brief introduction to rings.

Definition (Ring) Let R be a a set, and let $+, \cdot$ be two binary operations on R. We say that the triplet $(R, +, \cdot)$, often as R, is a ring if and only if the following conditions hold.

1. For all $a, b \in R, a + b = b + a$.
2. $a + (b + c) = (a + b) + c, \forall a, b, c, \in R$.
3. There exist a neutral element for $+$; that is, $\exists 0 \in R$ s.t. $a+0 = a, \forall a \in R$.
4. Each element has an additive oppposite, i.e., $\forall a \in R, \exists -a \in R$ such that $a + (-a) = (-a) + a = 0$.
5. $\cdot$ is associative; that is, $a(bc) = (ab)c, \forall a, b, c \in R$.
6. For all $a, b, c \in R, a(b + c) = ab + ac$.

Definition (Unital ring) Let $(R, +, \cdot)$ be a ring. We say that R is a unital ring, or a ring with unity, if there exists $1 \in R$ such that $1 \neq 0$ and $1a = a1 = a, \forall a \in R$.

Definition (Abelian ring) Let $(R, +, \cdot)$ be a ring. If for all $a, b \in R$ it holds that $ab = ba$; R is called commutative, or abelian, ring.

Definition (Integral domain) Let $(R, +, \cdot)$ be a commutative ring. If, for all $a, b \in R$ such that $ab = 0$, either $a = 0$ or $b = 0$; we say that R is an integral domain.

Definition (Division ring) Let $(R, +, \cdot)$ be a unital ring. We say that R is a division ring if, for all $a \in R$ exists a unique $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$.

Definition. Let $(R, +, \cdot)$ be a ring. We say that R is a field, if and only if R is a commutative division ring.

In the following we denote fields with $\mathbb{F}$, while Galois fields, i.e., fields with finitely many elements, with $\mathrm{GF}(q)$ where q is the number of elements in the fields.

In *Section 6.3* we discuss several codes, most of which make use of polynomial. In this context, an issue may occur: the chosen polynomial has no solution within the field in use. To solve it, we make use of field extensions, which are formally defined below.

Definition. Let $\mathbb{F}$ be a field. Denote by $\mathbb{F}[x]$ the set of polynomials with coefficients in $\mathbb{F}$[1]. Let $f(x) \in \mathbb{F}[x]$ be a irreducible polynomial, i.e., a polynomial with no solution[2]. We call field extension a field $\mathbb{F}'[x]$ in which we impose the existence of a solution for $f(x)$. That is,

$$\mathbb{F}'[x] = \{a + bi \mid a, b \in \mathbb{F} \text{ and } i \text{ root for } f(x)\}.$$

- 2.2 - Notions of projective geometry.

In what follows we assume that notions such as points, lines, planes in the geometrical sense, in addition to the notion of vector spaces are known to the reader.

Definition (Ray) Let $\mathbb{F}$ be a field, and let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Let $\mathbf{v} \in \mathcal{V}$ be non-zero. We call the set

$$[\mathbf{v}] = \{k\mathbf{v} \mid k \in \mathbb{F}^*\}$$

a ray.

Definition (Projective plane) Let $\mathbb{F}$ be a field and let $\mathcal{V}$ be a vector space over F. Let $\mathrm{Dim}(\mathcal{V}) = 3$. We define the projective plane $\mathbb{P}^2$ as the set of rays of V. An element $X \in \mathbb{P}^2$ is called a point.

[1] It can be proved that $\mathbb{F}[x]$ is also a field.

[2] To be more precise a polynomial with no solution is garanteed to be irreducible if its degree is at most 3.

In our discussion we usually consider projective points in *homogeneous* coordinates; that is, if $X = [x : y : z]$ is a projective point, at least one of x, y, z is non-zero.

The relation with ordinary plane geometry is done as follow:

- Any point $(x, y) \in \mathbb{R}^2$ becomes a projective point $[x : y : 1]$; and
- Any point $[x : y : 0] \in \mathbb{P}^2$, which do not belong to Euclidean geometry, are called point at infinity, and we denote them as $\mathcal{O}$.

We now present *elliptic curves* which are the main reason we discuss projective geometry.

Definition (Elliptic curves) Let $\mathcal{V}$ be a vector space over a field F. An elliptic curve $\mathcal{C}$ is the set of points of $\mathbb{P}^2$, that are solution to an equation of the form

$$y^2 = x^3 + ax + b$$

with $a, b \in \mathbb{F}$.

In our treating, we consider non-singular elliptic curves; that is, such that the discriminant is non zero, i.e., $\Delta = -16(4a^3 + 27b \neq 0)$.

In *Section* **??** we treat the ElGamal algorithm by means of elliptic curves; in this context given $\mathcal{C}$ an elliptic curve, we are required to compute the sum of two point $P, Q \in \mathcal{C}$, which is non trivial. In what remain of this section, we shall present how to do such addition.

Let $\mathcal{C}$ be an ellitptic over the field $\mathbb{F}$ and let $P = (x_1, y_1), Q = (x_2, y_2) \in \mathcal{C}$. Then, the following cases hold:

- The points are opposite of one another, then $P + Q = P + \mathcal{O} = P$.
- The two points are distinct; then we consider the line that passes through them. That is, we compute

  $$x_3 = m^2 - x_1 - x_2 \qquad y_3 = m(x_1 - x_3) - y_1$$

  where $m = {}^{(y_2 - y_1)}/_{(x_2 - x_1)}$ is the slope.
- The two point coincide; then we consider the tangent line. That is, we compute

  $$m = \frac{3x_1^2 + a}{2y_1}$$

  and x_3, y_3 as before.

- 3 - Public-key cryptography.

Modern days cryptography is based on a set of protocols based on the concept of *public-key* exchange. We proceed as follow: the idea of public-key cryptography is defined first, then some protocols will be presented. For the latter we discuss both the algorithms and the mathematical hard problem these are based on.

- 3.1 - The public-key exchange protocol.

Proposed for the first time in [1], it was a breaktrhough in cryptography. The idea is that of crypting the message as some discrete value, by means of a one-way function; that is, a function that is easy to compute, but whose inverse is hard to compute.

The whole cryptosystem relies on the assumption that one-way functions do exists; through the years several have been proposed and are used in schemes we discuss in the following, yet these are not garanteed to be one-way[1].

- 3.2 - The RSA algorithm.

The RSA algorithm, from the name of its inventors: Rivest, Shamir and Adleman (see [3] for major details); is a public-key algorithm based on the hard problem of integers factorization.

Formally speaking, the Integers Factorization Problem (IFP) is defined as follow: given a positive integer n, find its prime factors; that is, find $p_1, p_2, \ldots, p_k$ such that $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, with $\alpha_i \geq 1$.

- 3.2.1 - The algorithm.

Let m be the message to transmit; to cypher it by the RSA protocol do as follow: begin by choosing two odd distinc prime numbers p, q randomly, such that their bitlength is approximately the same. Compute $n = pq$ and $\lambda = (p-1)(q-1)$. Let $\mathcal{B}$ choose an integer e such that $\mathrm{GCD}(e, \lambda) = 1$; the pair (n, e) will be $\mathcal{B}$'s *public key*. Let d be the integer less then λ, such that $ed \equiv 1 \pmod{\lambda}$; this will be $\mathcal{B}$'s *private key*.

To encode the message $\mathcal{A}$ computes

$$c \equiv m^e \pmod{n}$$

and sends c to $\mathcal{B}$. Upon reciving the chypertext, $\mathcal{B}$ can easily decode m as

$$m \equiv c^d \pmod{n}. \tag{1}$$

[1] If one proves the existence of one-way functions, at the same time would solve the $\mathcal{P} = \mathcal{N}P$ problem.

To see why *Equation* (1) holds observe the following: since $ed \equiv 1 \pmod{\lambda}$, there exists k such that $ed = 1 + k\lambda$.

Now, we have that:

- If $\mathrm{GCD}(m, p) = 1$ by Fermat's little theorem
$$m^{p-1} \equiv 1 \pmod{p}.$$
Hence, raising both sides by $k(q-1)$ and multipling by m yields
$$m^{1+k\lambda} \equiv m \pmod{p} \implies m^{ed} \equiv m \pmod{p}.$$
- If $\mathrm{GCD}(m, p) = p$ the above still holds, as both sides will be congruent to 0.

Applying the same reasoning for q, since p and q are distinc, we conclude that
$$c^d \equiv m \pmod{n}.$$

Let us briefly overview why $\mathcal{E}$ cannot, at least no so easily, dechypher the message sent by $\mathcal{A}$. By the protocol both n, e are know to everyone, but the values of λ and d are not as these depend on p and q. Hence, the only choice left to $\mathcal{E}$ is to factor n; which we know is hard to do.

Example

Let $p = 37, q = 39$. We have $n = 1443$ and $\lambda = 1368$. Say $\mathcal{B}$ chooses $e = 263$ for is public key. At this point the pair $(n, e) = (1443, 263)$ is know to everybody. $\mathcal{A}$ converts is message into an integer $0 < m < n$, say $m = 1029$ and computes
$$c \equiv m^e \pmod{n} \implies 1320 \equiv 1029^{263} \pmod{1443}$$
and sends $c = 1320$ to $\mathcal{B}$. In the meanwhile, $\mathcal{B}$ computes d such that $eq \equiv 1 \pmod{\lambda}$, in our case $d = 671$. Hence, upon reciving $c, \mathcal{B}$ dechyphers it as
$$c^d \pmod{n} = 1320^{671} \pmod{1368} = 1029.$$

■

- 3.3 - The ElGamal algorithm.

After Diffie and Hellman published their paper, apart from the RSA protocol, a new scheme was proposed: the ElGamal protocol. Formally speaking, the algorithm is based on the Discrete Logarithm Problem (DLP): fixed a group $\mathrm{GF}(p)$ and two non-zero elements g and h, find x such that
$$g^x \equiv h \pmod{p}.$$

We discuss the algorithm as follow: first we describe the algorithm in the case of integers modulo a prime, then we discuss it in the context of elliptic curves.

- 3.3.1 - The algorithm: the modulo p case.

Let a Trusted Third Party (TTP) choose a (large) prime p; consider the field $\mathrm{GF}(p)^*$ and let γ be a unit. That is, let γ be such that $\langle\gamma\rangle = \mathrm{GF}(p)^*$. Tell $\mathcal{A}$ to choose $a \in \{2, \ldots, p-1\}$ and compute

$$\alpha \equiv \gamma^a \pmod{p}.$$

The quantity α is public in the channel.

Let m be a message $\mathcal{B}$ wants to send to $\mathcal{A}$. We can assume m to be an integer in the range $[2, p]$. To chyper $m, \mathcal{B}$ chooses a integer k, computes

$$c_1 \equiv \gamma^k \pmod{p} \quad \text{and} \quad c_2 \equiv m\alpha^k \pmod{p}$$

and send the pair (c_1, c_2) to $\mathcal{A}$.

Observe that $\mathcal{A}$ can easily decode the message; in fact, $\mathcal{A}$ can compute

$$x \equiv c_1^a \pmod{p}$$

hence, x^{-1} is known as well. Once x^{-1} is known, computing $x^{-1}c_2 \pmod{p}$ gives us m. It's easily seen that

$$\begin{aligned} x^{-1}c_2 &\equiv (c_1^a)c_2 \pmod{p} \\ &\equiv (\gamma^{ak})^{-1}m\alpha^k \pmod{p} \\ &\equiv (\gamma^{ak})^{-1}m\gamma^{ak} \pmod{p} \\ &\equiv m \pmod{p}. \end{aligned}$$

Example

Say out TTP chose $p = 47$ and $\gamma = 3$. To compute their public key $\alpha = \gamma^a, \mathcal{A}$ chooses $a = 27$. Hence, the value $\alpha = 34$ is known on the channel.

Let $m = 45$ be the message $\mathcal{B}$ wants to send to $\mathcal{A}$, and to compute c_1, c_2 they choose $k = 11$. Then, we have

$$c_1 \equiv 4 \pmod{p} \qquad c_2 \equiv 5 \pmod{p}.$$

Upon recieving the pair (c_1, c_2) can easely decode m. Observe that, $x \equiv c_1^a \equiv 21 \pmod{47}$ and it can be easely checked that $x^{-1} = 9$. Therefore, it follows

$$\begin{aligned} x^{-1}c_2 &\equiv \\ &\equiv c_1^a c_2 \\ &\vdots \\ &\equiv 45 \pmod{47} \end{aligned}$$

which is m as expected. ■

- 3.3.2 - The algorithm: the elliptic curve case.

A trickier implementation of the ElGamal algorithm makes use of elliptic curves. Recall that the "sum" of two points P and Q of some elliptic curve $\mathcal{C}$ is not trivial.

Back to the algorithm: a TTP fixes a elliptic curve $\mathcal{C}$ over the some field $\mathrm{GF}(p)$. Analogously to the previous case, an element $\gamma \in \mathcal{C}$ is fixed and the quantitie $\alpha = a\gamma$ is computed and published over the channel. The remainder of the algorithm is similar to that previously discussed; the main difference is that multiplications are replaced by additions.

Remark. Why do we prefer elliptic curves then? The answer is simple: the computation of $n\gamma$ is expensive.

- 4 - Digital signature.

In todays era of digital documents, it is fundamental to have a way to garantee the ownership of a file, a program or whatsoever. Is in this context that digital signature found its use.

In what follows, we will denote by $\mathcal{S}$ the signature of the document and by $\mathcal{V}$ the trusted entity that proceed to verify the signature; addionally, what we consider is not the document in itself, rather its hash.

- 4.1 - RSA digital signature.

Signing a document by means of the RSA algorithm is somewhat straightforawrd. Let d is the hash value of some document $\mathcal{A}$ produced; to sign d she proceed as follow: she chooses two primes p and q, computes their product $n = pq$ and publish n and a public verification exponent.

Following a similar reasoning to that of *Section 3.2.1*, the congruence

$$sv \equiv 1 \pmod{(p-1)(q-1)}$$

is solved with respect to s. Here s is the *signing exponent* and v the *verification exponent*. Lastly, the document is singned as

$$\mathcal{S} \equiv d^s \pmod{n}.$$

To check the signature, $\mathcal{V}$ simply computes

$$\mathcal{S}^v \pmod{n}$$

and checks whether this is d or not. The correctness of this is given by Euler's formula.

Similarly to the RSA encryption scheme, the RSA signature securety follows from the hardness of computing the v-th root of d, rather then the factorization of n.

- 4.2 - ElGamal digital signature.

Let a TTP choose a large prime p and an integer g such that $\mathrm{GF}(p) = \langle g \rangle$. Say $\mathcal{A}$ has to sign a document d; before doing so they must choose a signing exponent s and compute the quantity

$$v \equiv g^s \pmod{p}.$$

The quantity v, alongside the parameter g and p, represent $\mathcal{A}$'s public verification key.

The signature itself is produced, by choosing an integer $e < p$ and computing the quantities

$$S_1 \equiv g^e \pmod{p} \qquad S_2 \equiv (d - sS_1)e^{-1} \pmod{p-1}.$$

To verify the signature, $\mathcal{V}$ checks that

$$v^{S_1} S_1^{S_2} \pmod{p} = g^d \pmod{p}.$$

Note that the above equation correctly checks the signature; in fact, we have that

$$\begin{aligned} v^{S_1} S_1^{S_2} &\equiv g^{sS_1} g^{eS_2} \\ &\equiv g^{sS_1 + eS_2} \equiv g^{sS_1 + e(d - sS_1)e^{-1}} \equiv g^d \pmod{p}. \end{aligned}$$

- 5 - Integers factorization.

In this section we describe some interesting algorithms to factor integers; additionally, we describe how Euler algorithm to compute the MCD can be improved and how to efficiently compute the modulo of a large integer.

- 5.1 - Pollard's $p-1$ algorithm.

Should be clear by now that prime numbers are at the core of cryptography, since these are hard to find[2]. On the other hand, as in the case of RSA, we need to find distinct prime numbers that are not big, yet not easily computable. Meaning a test for primality would be a good tool. Luckly such tool exists – Pollard's $p-1$ algorithm – though it's applicable to a specific type of numbers.

Let's consider the details. Pollard's algorithm is based on the concept of B-smooth numbers; that is, numbers whose prime factors are all smaller than or equal to a given bound B. Precisely, let $n = pq$, where p, q are prime, be the integer to factorize; assume we have found a integer L such that

$$p-1|L \text{ and } q-1 \not| \, L.$$

This means that there exists i, j and $k, k \neq 0$ such that

$$L = i(p-1) \text{ and } L = j(q-1) + k.$$

By Fermat little theorem we have

$$\begin{aligned} a^L &\equiv 1 \pmod{p} \\ a^L &\equiv a^k \pmod{q}. \end{aligned}$$

Then, we have that

$$p|a^L - 1 \text{ and } q \not| \, a^L - 1,$$

for most values of a. It follows that $p = \mathrm{GCD}\big(a^L - 1, n\big)$.

The interesting part is that, if $p-1$ is such that it's the product of small prime numbers, then it divides $n!$. In other words, what we do is fix $a \in \mathbb{Z}^+$ and compute $\mathrm{GCD}\big(a^{n!}, n\big) = d$; if $d \neq 1$ or $d \neq n$ we have found a non-trivial factor, otherwise we try again by changing the value of a.

Is easily noted that the algorithm has, at least apparently two severe drawbacks: the computation of the factorial and the consequential complexity. Let us say that, how would be clear later in this section, such drawbacks do not exists.

Back to the algorithm: we've said that Pollard's algorithm works if either $p-1$ or $q-1$ are product of small primes; the question is how likely this

[2]The statement is true unless *Riemann* hypothesis turns out to be true.

happens? It's here that B-smoothness plays a key role: if n is B-smooth, then it divides $B!$; additionally, any prime l dividing n has to be less than or equal to B.

- 5.2 - Pollard's ρ algorithm.

We now discuss an algorithm that can be used to solve, when it works, the DLP. Hence, this algorithm is a good tool for cracking ElGamal-based cryptosystems.

The idea is the following: let S be the set of elements we are working with, and consider a function $f : S \to S$ that randomly distributes the elements in S. Since for our purpouse S is finite, by applying f repetedly it will happen that

$$f(x_i) = f(x_j), i \neq j.$$

The only problem is to determine such a function.

In our context of the ElGamal algorithm, let GF(p) be the choosen field and let g be a primitive root. Then, as prosed in ([2]), we can use the function

$$f(x) = \begin{cases} gx, \text{if } 0 \leq x < p/3; \\ x^2, \text{if } p/3 \leq x < 2p/3; \\ ax, \text{if } 2p/3 \leq x < p; \end{cases}$$

where x is reduced modulo p. We can assume that $x_0 = 1$ and by applying f we get simply

$$g^{\alpha_i} a^{\beta_i}$$

where the values of α_i, β_i are given by

$$\alpha_i = \begin{cases} \alpha_{i-1} + 1, \text{if } 0 \leq x < p/3; \\ 2\alpha_{i-1}, \text{if } p/3 \leq x < 2p/3; \\ \alpha_{i-1}, \text{if } 2p/3 \leq x < p; \end{cases}$$

$$\beta_i = \begin{cases} \beta_{i-1}, \text{if } 0 \leq x < p/3; \\ 2\beta_{i-1}, \text{if } p/3 \leq x < 2p/3; \\ \beta_{i-1} + 1, \text{if } 2p/3 \leq x < p. \end{cases} .$$

Then, if we compute $y_i = f(f(y_{i-1}))$, with $y_0 = x_0$ and check when $y_i = x_i$ we get something of the form

$$g^{\alpha_i} a^{\beta_i} = g^{\gamma_i} a^{\delta_i}$$

where γ_i, δ_i are defined by the same relation of α and β. Hence, we have that

$$g^{\alpha_i - \gamma_i} = a^{\beta_i - \delta_i}.$$

Putting $u = \alpha_i - \gamma_i$ and $v = \beta_i - \gamma_i$ and considering the logarithm, we get

$$v \log_g a \equiv u \pmod{(p-1)}.$$

Recall that p is known, therefore, if $\mathrm{GCD}(v, p-1) = 1$ we can multiply by v^{-1} and easely compute the logarithm. Viceversa, if $\mathrm{GCD}(v, p-1) \geq 2$, we can apply the Eucledian algorithm to find s such that $su \equiv d \pmod{(p-1)}$. Multiplying both sides by s, we get

$$d \log_g a \equiv su \pmod{(p-1)},$$

whose solutions is the set

$$\left\{ \frac{w}{d} + k\frac{p-1}{d} \,\middle|\, 0 \leq k < d \right\}.$$

- 5.3 - Lenstra factorization.

Differently from the two algorithms previously described, Lenstra's factorization algorithm assumes we are working with an elliptic curve $C*$ over the ring $\mathbb{Z}/n\mathbb{Z}$. The key idea is that, while computing kP, for some $k \in \mathbb{Z}$ and $P \in \mathcal{C}$, we will eventually have to compute the inverse of a element of the ring that admtis no inverse. Precisely we proceed as follow:

1. Let $\mathcal{C}$ be the elliptic curve $Y^2 = X^3 + AX + B$ over the ring $\mathbb{Z}/n\mathbb{Z}$.
2. Fix a point $P = (a, b)$ at random and $B \equiv b^2 - a^3 - aA \pmod{n}$.
3. For $j = 2, 3, 4, \ldots$ up to some bound do:
   (a) Compute $Q \equiv jP \pmod{N}$ and set $P = Q$.
   (b) If the previous step fails, we have found a non trivial divisor d.
   (c) If $d \leq n$, return d.
   (d) If $d = n$ go back a choose a new point, a new curve or both.
4. Go back to step 2.

Lenstra's factorization algorithm requires $\mathcal{O}\left(e^{\sqrt{(\log n)(\log\log n)}}\right)$ time.

- 5.4 - Extended Euclidean algorithm.

Assume we are interested into computing the GCD d of two integers a and b. From what we have discussed so far, should be obvious by now, that we are also interested into knowing what are the coefficient α, β such that

$$d = \alpha a + \beta b.$$

Hence, the classic Euclidean algorithm cannot be used. Thankfully, there exists an *extended* algorithm that solves the problem. We present below the pseudocode.

Algorithm 1: Extended Euclidean Algorithm

Input: two non-negative Integers a, b with $a \geq b$.
Output: $d = \text{GCD}(a, b)$ and the integers x, y such that $ax + by = d$.

If $b = 0$, return $(a, 1, 0)$.
Set $x_2 \leftarrow 1, x_1 \leftarrow 0, y_2 \leftarrow 0, y_1 \leftarrow 1$.
While $b > 0$ do:
    $q \leftarrow \lfloor a/b \rfloor, a \leftarrow a - qb, x \leftarrow x_2 - qx_1, y \leftarrow y_2 - qx_1$.
    $a \leftarrow b, b \leftarrow r, x_2 \leftarrow x_1 x_1 \leftarrow x, y_2 \leftarrow y_1,$ and $y_1 \leftarrow y$.
Return (a, x_2, y_2).

- 5.5 - The Square and multiply algorithm.

We have discussed so far of how the securety of our algorithms is given by, among other things, the necessifty of computing big numbers reduced in modulo. The question then is, can we ease the task when the user has rights on the data? The affirmative answer is provided by the *Square-and-Multiply* algorithm. *Figure 2* shows the pseudo-code of the algorithm.

Informaly speaking, given a number in exponential form; we decompose the exponent as subsequent powers of two, compute the reduced exponentiations and multiply the results, reducing in modulo accordingly.

Algorithm 2: Square and Multiply algorithm

Input: $g \in \mathbb{Z}_n$ and an integer $0 \leq e < n$, whose binary representation is $e = \sum_{i=0}^{t} e_i 2^i$
Output: $g^e \pmod{n}$

Set $b \leftarrow 1$. If $e = 0$ return b.
Set $A \leftarrow g$.
If $e_0 = 1$, set $b \leftarrow g$.
For i from 1 to t do:
    Set $A \leftarrow A^2 \pmod{n}$.
    If $e_i = 1$, set $b \leftarrow A * b \pmod{n}$.
Return b.

More rigorously, let g be the base number, let e be the exponent and n the modulo. That is, we want to compute

$$g^e \pmod{n}.$$

For the sake of our discussion, assume $e >> 1$.

Essentially, we consider all the subsequent powers of 2, up to $2^{\log e}$. Write e in it's binary expansion and compute g^e as the product of g^{2^i}, where 2^i is in the binary expansion of e.

- 6 - Code theory.

Alongside information theory, code theory is at the core of modern CS. To give the reader an idea, the former, due to *C. Shannon*, deals with how the communication among systems works, providing theoretical tools to solve/mitigate several issues (eg. noise). On the other hand, code theory, due to *R. W. Hamming*, deals with how informations are carried inside the channel.

In this section we focus on some specific class of codes, describing how these are obtained and what are their propertes.

Let us note that we assume the reader to have some basic knowledge of linear algebra; precisely, we assume the reader to know what vector spaces are, how to sum vectors, and how to work with matrixes.

- 6.1 - Basic definitions.

Definition. A finite set of symbols A is said to be an alphabet, if its elements are used to represet strings.

Definition. Let A be an alphabet. We define word (or string) an n-tuple of A^n. That is,

$$A^n = \{\mathbf{w} = (x_1, \dots, x_n) \mid x_i \in A, \forall i\}.$$

Additionally, given a word $\mathbf{w}$ its weight $\|\mathbf{w}\|$ is the number of non-zero components in $\mathbf{w}$.

From the above definition two remarks can be made about the weight of a word:

1. For any alphabet $A, \mathbf{w} \in A^n$ has weight 0 if and only if $\mathbf{w} = \mathbf{0}$.
2. For any two strings $\mathbf{w}_1, \mathbf{w}_2 \in A^n$, it holds

$$\|\mathbf{w}_1 - \mathbf{w}_2\| = 0 \iff \mathbf{w}_1 = \mathbf{w}_2.$$

Definition (Hamming's distance) Let $\mathbf{w}_1, \mathbf{w}_2 \in A^n$, for some alphabet A. We define the (Hamming) distance of $\mathbf{w}_1, \mathbf{w}_2$ to be the number of components for which these differ.

Remark. It can be easely proved that the Hamming's distance is a metric.

Definition (Code) Let A be an alphabet, and consider the vector space A^n these creates. We define code a subset $\mathcal{C}$ of A^n.

Definition (Minimal distance) Let $\mathcal{C}$ be a code. We say that $C*$ has Minimal distance d if

$$d = \min\{|\mathbf{w}| \mid \mathbf{w} \in \mathcal{C}\}.$$

That is, the minimal distance of a code is the minimal weight of all the words in $*$.

In our discussion of codes we assume A to be always some finite field $\mathbb{F}$, or equivalently a Galoi field of q elements $\mathrm{GF}(q)$.

- 6.2 - Linear codes.

Definition. Let $\mathcal{C} \subseteq A^n$ be a code. If we have that $\mathcal{C}$ is a vector subspace of A^n, we say that $\mathcal{C}$ is linear. That is, $\mathcal{C}$ is a linear code if:

1. For any $\mathbf{c}_1, \mathbf{c}_2 \in \mathcal{C}, \mathbf{c}_1 + \mathbf{c}_2 \in C$; and
2. For any $\mathbf{c} \in \mathcal{C}, \forall \alpha \in A, \alpha\mathbf{c} \in \mathcal{C}$.

Each code $\mathcal{C}$ is identified by the tuple (n, k, d), where n is the size of the words (i.e. the number of components), k the dimesions of $\mathcal{C}$ and d the minimal distance. In the case of linear codes we write $[n, k, d]$ instead of (n, k, d).

The question now is: how do we can represent a linear code? Since $\mathcal{C}$ is a vector subspace, it must have a basis of vectors that generates the whole subspace. If $\mathbf{c}_1, \ldots, \mathbf{c}_k$ form a basis for $\mathcal{C}$, we can denote by

$$G = \begin{pmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_k \end{pmatrix}$$

the *generator matrix* of $\mathcal{C}$. It can be easely seen that any word in $\mathcal{C}$ is given by the linear combination of rows in G.

Alternatively, we can make use of a *parity-check matrix*. That is an $r \times n$ matrix H over A, such that

$$\mathbf{c} \in \mathcal{C} \iff H\mathbf{c}^T = \mathbf{0}.$$

Note that $r = \text{Rank}(H) = n - k$, from the dimesional relationship.

- 6.2.1 - Theorems and inequalities.

We now present some results that are at the core of the codes we present later. Though we try to be as formal as possible, we also try to make the proof undestandable to those with little-to-no mathematical background.

Theorem 6.1 *Let $\mathcal{C}$ be linear code defined by a parity matrix H and let d its minimal distance. Then,*

1. *Every d columns of H are linearly dependent, and*
2. *Every $d - 1$ columns of H are linear independent.*

Proof

1. Recall that the minimal distance of a code, is also the minimal weight of distinct words in it. It follows that

$$\exists \mathbf{w} \in \mathcal{C} \text{ s.t. } \mathbf{w} = (0, \ldots, 0, \alpha_1, 0, \ldots, 0, \ldots, 0, \alpha_d, 0, \ldots, 0)$$

where $\alpha_1, \ldots, \alpha_d \neq 0$. By hypotesis $H\mathbf{w}^t = \mathbf{0}$, hence there exist linear combination of columns of H that is the zero-vector. Thus, we have d linearly dependent columns.

2. Trivial. By contradiction, if $d-1$ columns would be linearly dependent then by 1., there exists a word $\mathbf{w} \in \mathcal{C}$ with $d-1$ non zero components such that $H\mathbf{w}^t = \mathbf{0}$. But then the minimal weight of $\mathbf{w}$ is $d-1$ and not d as supposed. ■

Theorem 6.2 *Let $\mathcal{C}$ be a linear code and let d be its minimal distance. Then,*

1. *The code can detect up to d errors, and*
2. *Up to $(d-1)/2$ errors can be corrected.*

Proof

1. Let $\mathbf{w} \in \mathcal{C}$ be the word transmitted, and let $\mathbf{z}$ be the one recived. Recall that both, sender and reciver, know the code. Assume that during the transmission e errors occured. If $e < d$, then $\mathbf{z}$ is not a valid word, in fact we would have $d(\mathbf{w}, \mathbf{z}) = e < d$. On the other hand, if $e > d$, we can say nothing about $\mathbf{z}$ and it would be accepted as valid.

2. By contradiction, suppose that $\mathbf{w}_1 \neq \mathbf{w}_2$ are two words with distance at most $(d-1)/2$ from $\mathbf{z}$. That is,

$$d(\mathbf{w}_1, \mathbf{z}) \leq \frac{d-1}{2}, \qquad d(\mathbf{w}_2, \mathbf{z}) \leq \frac{d-1}{2}.$$

By the trinangular inequality, we have

$$d(\mathbf{w}_1, \mathbf{w}_2) \leq d(\mathbf{w}_1, \mathbf{z}) + d(\mathbf{w}_2, \mathbf{z}) = \frac{d-1}{2} + \frac{d-1}{2} = d-1,$$

but, by hypotesis, $\mathbf{w}_1, \mathbf{w}_2$ have distance d. Meaning that $\mathbf{w}_1 = \mathbf{w}_2$. ■

Theorem 6.3 (Singleton bound) *Let $\mathcal{C}$ be a $[n, k, d]_q$ linear code defined by a parity matrix H. Then, it holds*

$$d \leq n - k + 1.$$

Proof Recall that by the dimensional relation $r = \mathrm{Rank}(H) = n - k$. By *Theorem 6.1* $d-1$ columns of H are linearly independent. Combining the two, we have

$$d - 1 \leq r \iff d - 1 \leq n - k \iff d \leq n - k + 1.$$

■

A code that satisfies the Singleton bound is said to be optimal.

Theorem 6.4 (Hamming inequality) *Let $\mathcal{C}$ be a $[n, k, d]_q$ linear code. Then, for any transmission errors $e \leq {}^{(d-1)}/_2$, it holds*

$$q^e \left(\sum_{i=0}^{e} (q-i)^i \binom{n}{i} \right) \leq q^n.$$

Proof Trivial. Let $\mathbf{w}$ be the word transmitted, we have that:

- If $e = 0$, we recived the correct word.
- If $e = 1$, we have to consider the wordis with distance 1 from $\mathbf{w}$, which are exactly $(q-1)n$.
- If $e = m$, with $m \leq {}^{(d-1)}/_2$, we have $(q-m)^m \binom{n}{m}$ word at distance m from $\mathbf{w}$.

Hence summing all up, we get

$$q^e \left(\sum_{i=0}^{e} (q-i)^i \binom{n}{i} \right) \leq q^n.$$

■

A code satisfying Hamming inequality is said to be perfect.

- 6.2.2 - Hamming codes.

Hamming code, proposed by Richard W. Hamming, is one of the most used linear codes to this day. More precisely, Hamming proposed a family of codes all with their minimal distance set to three. It also turns out that this family of codes is perfect.

Before presenting the generalized Hamming code, we describe two of its most used variants: the $[7, 4, 3]_2$ and the $[3, 1, 3]_2$ Hamming code.

$[7, 4, 3]_2$ Hamming code.

The code is composed of 16 strings of length 7 over GF(2). Hence, apart from isomorphism, the generating matrix and the parity one are the one below.

$$G = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \qquad H = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

$[3,1,3]_2$ **Hamming code.**

The code is composed of 2 strings of length 3 over GF(2). Hence, apart from isomorphism, the generating matrix and the parity one are the one below.

$$G = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \qquad H = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

In addition to being perfect, this code is also optimal as it satisfies the singleton bound.

Generalized Hamming codes.

Consider a field $\mathbb{F} = \mathrm{GF}(q), q = p^m$. Fix the vector space $\mathbb{F}^r, r \geq 1$. Here, each line vector $\langle \mathbf{w} \rangle$ is a non-empty string $\mathbf{w}$ and all it's multiples. If we consider the matrix made up by all the $\mathbf{w}$, we get a matrix H with dimension

$$r \times \frac{q^r - 1}{q - 1}.$$

A code $\mathcal{C}$ with this H as parity matrix is called generalized Hamming code, and it's parameter n, k, d are

$$n = \frac{q^r - 1}{q - 1}, \quad k = \frac{q^r - 1}{q - 1} - r, \quad d = 3.$$

- 6.2.3 - Hadamard codes.

Named after *Jaques Hadamard*, was actually proposed by *R. C. Bose* and *S. S. Shrikhande*, and refers to the family of $[2^k, k, 2^{k-1}]$ linar code. We can distinguish two "classes" of Hadamard codes: the plain one and the augmented one. The former have a message length of $k = m$; the latter of $k = m + 1$. In what follows, we refer to the second class of codes as augmented Hadamard codes.

To construct an Hadamard code, fix GF(2) as field and let k be the lenght of the words. Then, the generator matrix for the code is composed of all binary strings of lenght k, lexicographically ordered. That is, the matrix

$$G = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ y_1 & y_2 & \cdots & y_{2^k} \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}$$

where $y_i \in \{0,1\}^k$.

With the same assumptions, the generator matrix for an augmented Hadamard code is composed by all the binary strings of lenght k whose first component, is a 1.

- 6.3 - Cyclic codes.

Definition. Let $\mathcal{C} \subseteq A^n$ be a code. If $\forall \mathbf{w} \in \mathcal{C}$, the cyclic shift of $\mathbf{w}$ is still a word of the code, we say that $\mathcal{C}$ is cyclic.

Remark. Algebraically speaking, we say that $\mathcal{C} \subseteq A^n$ is cyclic if and only if $\mathcal{C}$ is an ideal of A^n.

In our discussion of cyclic codes we treat words in a code as polynomials. That is, if $\mathbf{w} = (w_0, w_1, \ldots, w_{k-1})$ is a word of some cyclic code, then it refers to the polynomial $f(x) = w_0 + w_1 x + \ldots + w_{k-1}x^{k-1}$.

As for the case of linear codes, to each cyclic code we can associate two polynomials: one that generates the code, call it $g(x)$, and one for code checking. It can be proved that $g(x)$ is minimal and has degree $n-k$, additionally it divedes the polynomial $x^n - 1$.

That is, a cyclic code $\mathcal{C}$ defines the set

$$\langle g(x) \rangle = \{f(x) \in \mathbb{F}_n[x] \mid f(x) = g(x)u(x), u(x) \in \mathbb{F}_{n-\deg g}[x]\}.$$

It should be noted that a cyclic shift in this context corresponds to a multiplication by x. In fact, if $\mathbf{w} = (w_0, w_1, \ldots, w_{k-1})$ is a word, then $(\mathbf{w}' = (w_{k-1}, w_0, \ldots, w_{k-2})$ is also a word. Hence, using the same matrix notation as for linear codes, the generator matrix for the code $\mathcal{C}$, with generator $g(x)$, is

$$G = \begin{pmatrix} g_0 & g_1 & \cdots & g_{n-k} & & & \\ & g_0 & g_1 & \cdots & g_{n-k} & & \\ & & \ddots & \ddots & \cdots & \ddots & \\ & & & g_0 & g_1 & \cdots & g_{n-k} \end{pmatrix}$$

On the other hand, to understand how the parity matrix for a cyclic code is obtained, we have to consider the following. Since $g(x)|x^n - 1$, and $\deg g = n-k$, there must exist a polynomial $h(x)$ such that

$$g(x)h(x) = x^n - 1 \wedge \deg h = k.$$

Though we don't descibe them, it should be noted that $h(x)$ is the generating polynomial for the dual code[3]of $\mathcal{C}$.

Consider the following

$$\begin{aligned} \langle g(x) \rangle \ni c(x) &= f(x)g(x) = c_0 + c_1 x + c_2 x^2 + \ldots + c_{n-1}x^{n-1} \\ \langle h(x) \rangle \ni d(x) &= l(x)h(x) = d_0 + d_1 x + d_2 x^2 + \ldots + d_{n-1}x^{n-1} \end{aligned}$$

if we multiply them we get

$$c(x)d(x) = f(x)g(x)l(x)h(x) = m(x)(x^n - 1) = m(x)x^n - m(x).$$

Considering the degrees of f and l we have:

$$\deg f \leq n-1-\deg g \leq k-1 \wedge \deg l \leq n-1-\deg h \leq n-k-1$$

that is, $\deg m(x) \leq n-2$, meaning no term of degree $n-1$ exists. In other words, the sum

$$c_0 d_{n-1} + c_1 d_{n-2} + \cdots + c_{n-1} d_0 = 0;$$

equivalently, the product $(c_0, \ldots, c_{n-1}) * (d_{n-1}, \ldots, d_0)^t = 0$.

If we consider it in is polynomial form, the second term of the product is nothing else but the reverse of $d(x)$, which is obliviously generated by some polynomial $\tilde{h}(x)$ in turns reverse of $h(x)$.

Summing up, the matrix H generated by $\tilde{h}(x)$ is exactly the parity matrix for our code $\mathcal{C}$.

[3]Informally speaking, these can be thought as the complement of some code $\mathcal{C}$.

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Acronyms.

DLP Discrete Logarithm Problem. 7

GCD Greatest Common Divisor. 2

IFP Integers Factorization Problem. 6

LCM Least Common Multiple. 2

TTP Trusted Third Party. 8